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| Researcher |

| Evolutionary Geneticist |

Professional Summary

A **Ph.D** in Animal Genetics with extensive expertise in computational biology to support evolutionary research, focusing on analyzing large-scale datasets and developing scalable bioinformatics pipelines. Proven expertise in various omics analyses, including genomics, transcriptomics, proteomics, epigenetics, and data integration.

A researcher at the Swedish University of Agricultural Sciences (SLU) and the key bioinformatician to WP10 of the EU-funded DEFEND project (Horizon 2020), leading bioinformatics efforts to uncover host transcriptomic signatures associated with resilience mechanisms against two emerging viral infections.

Proficient in high-performance computing (HPC) environments, with expertise in developing reproducible workflows using Python, R, and Bash scripting; an applicable computational skillset that broadly translates across diverse domains, including animal, plant, and human research.

Skilled in statistical inference, including Bayesian modeling, and downstream analyses such as gene ontology (GO) enrichment and KEGG pathway analysis to extract meaningful insights from large datasets.

An experienced molecular geneticist with hands-on knowledge of wet-lab protocols, including DNA/RNA extraction, PCR/qPCR, and sequencing, and the ability to bridge laboratory insights with computational analyses.

Publications in high-impact journals demonstrate innovative approaches to address complex biological questions, mainly in evolutionary and population genomics. Passionate about collaborative science and contributing to interdisciplinary research environments that aim to model evolutionary complexity across species and ecosystems.

Education

Ph.D. in Animal Genetics | University of Tehran, IRAN | July 2016
Dissertation: Computational strategies for integration of transcriptomic data (RNA-Seq) into genomic-enabled predictions

M.Sc. in Animal Genetics | University of Tehran, IRAN | June 2002
Focus on the molecular phylogenetics of local animal populations.

Careers

Postdoctoral Fellow (Voluntary) | Latvian Institute of Organic Synthesis (LIOS), Latvia | Oct 2025 – Present
EU-funded TARGETWISE Project

- Analyzing differential gene and metabolite expression using transcriptomic and metabolomic data from knockout mice.
- Building dynamic phenotypic trajectory models using Generalized Additive Mixed Models (GAMMs).
- Applying machine learning methods for multi-omics integration and systems-level analyses.
- Developing deep learning pipelines for ancient DNA classification of ancient versus modern DNA without reliance on reference genomes using CNN-based models.

Postdoctoral Fellow (Remote) | North Dakota State University (NDSU), USA | Jun 2024 – Sep 2024

USDA-ARS-funded Project No. 3060-21000-047-015S: "Multi-Omics Data and Non-Coding Genomic Variants to Improve Crop Climate Resilience" under "Predictive Crop Performance Research Initiative" program

- Navigating Transposable Elements (TEs) flanking each syntenic gene across pan-genomes as the potential regulatory elements.

- Developed a large-scale software library for TE mapping across pan-genomes with over 47,000 gene cards, and calculated LTR Assembly Index (LAI) to evaluate genome assembly quality.

Researcher | Swedish University of Agricultural Sciences (SLU), Sweden | Jan 2022 – Dec 2023

- Led transcriptomics and genomic analyses within the EU-H2020 DEFEND project.
- Designed reproducible pipelines and mentored team members in bioinformatics workflows.

Assistant Professor | Animal Science Research Institute of Iran (ASRI), Iran | Aug 2016 - Jan 2022

- Led research in genomics, transcriptomics, and multi-omics data integration.
- National Coordinator for Animal Genetic Resources at FAO/DAD-IS (2019–2022).

Visiting Researcher | Swedish University of Agricultural Sciences (SLU), Sweden | Mar - Sep 2015

- Variant calling, Allele-Specific Expression (ASE), parent-of-origin effects, and personalizing the genome.

Research Associate | Animal Science Research Institute of Iran (ASRI), Iran | 2006 – 2016

- Contributed to research projects, data collection, and statistical analyses.

Laboratory Engineer | Animal Science Research Institute of Iran (ASRI), Iran | 2004 – 2006

- Supported molecular genetics laboratory work.

Analytical and Laboratory Skills

Genomics: Genome-Wide Association Studies (GWAS), haplotyping, genome-enabled prediction, selection signature, transposable element annotation, pan-genomics, and comparative genomics across taxa.

Transcriptomics: RNA-Seq analysis, differential gene expression (DGE), transcriptome-wide association studies (TWAS), eQTL mapping, allele-specific expression (ASE), and variant calling.

Epigenomics: DNA methylation analysis and epigenome-wide association studies (EWAS).

Statistical Modeling: Generalized linear models (GLM), Bayesian hierarchical modeling (JAGS), descriptive and inferential statistics, PCA, and temporal/spatial models.

AI Modeling: Supervised and unsupervised machine learning and deep learning for predictive inference, and integrative analysis across heterogeneous omics datasets.

Bioinformatics: Development of scalable, reproducible pipelines using Snakemake and Nextflow; version control via GitHub; cloud computing, high-performance computing (HPC); multi-omics data integration, data visualization, and simulation.

Programming: R, Python, Bash/Shell scripting under Linux/Unix-based environments.

Downstream Analysis: Gene annotation, gene ontology (GO), pathway analysis, gene set enrichment, and multi-layer networking.

Molecular Genetics: DNA/RNA extraction, conventional PCR and real-time PCR (qPCR), Sanger and NGS-based sequencing.

Collaborative Research: Experience in international, interdisciplinary collaborations; mentoring of graduate students; and active contribution to open and reproducible science.

Training & Teaching

Courses, recently completed: Reproducible Genomics Workflows (CRG, 2020) | RNA-Seq Data Analysis (SLU, 2022) | Bayesian Ecological Models (SLU, 2023) | DNA Methylation Profiling (SLU, 2024)

Teaching: Bioinformatics | Genome-enabled predictions | Transcriptomics | Genetic resistance

Awards and Recognition

Best Young Researcher Award, AREEO, Tehran, Iran – 2010
Best Researcher Award, ASRI, Karaj, Iran – 2007

Languages & Memberships

Languages: English: Fluent | Persian (Farsi): Native | Swedish: Basic

Memberships: Member, Swedish Association of University Teachers and Researchers (SULF), 2022-2024 | Board Member, Iranian Genetics Society (IGS), 2014-2016 | Board Member, Iranian Society of Animal Science (ISAS), 2019-2022

Selected Publications (the Persian papers and presentations have not been included)

| Scopus | Web of Science | SLUpub (2022 - 2025) | Google Scholar |

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910 Citations by 873 documents | 52 Documents | 12 h-index

Banabazi, M. H. et al. (2025). Host transcriptome profiling for resistance against Lumpy Skin Disease (LSD), BMC Research Notes, 18(1),299. DOI First author

Banabazi, M. H. et al. (2024). The transcriptomic insight into the differential susceptibility of African Swine Fever in inbred pigs. Nature Scientific Reports, 14, 5944. DOI First author

Mahmoodi, S. et al. (2024). Elucidating genetic variability between randomly bred domestic cats and Persian domestic cats from different geographical locations using microsatellite markers. Veterinary Medicine and Science, 10(6), e70004. DOI Corresponding author

Asadollahpour Nanaei, H. et al. (2024). High-throughput DNA sequence analysis elucidates novel insight into the genetic basis of adaptation in local sheep. Tropical Animal Health and Production, 56, 150. DOI Co-author

Vahedi, S. M. et al. (2023). Population genetic analysis and scans for adaptation and contemporary selection footprints provide genomic insight into *aus*, *indica* and *japonica* rice cultivars diversification. Journal of Genetics, 102, 43. DOI Co-author

Golshani Jourshari, M. et al. (2023). Genome-wide association study on abdomen depth, head width, hip width, and withers height in native cattle of Guilan (*Bos indicus*). PLOS ONE, 18(8), e0289612. DOI Corresponding author

Vahedi, S. M. et al. (2023). Epidemiology, pathogenesis, and diagnosis of Aleutian disease caused by Aleutian mink disease virus: a literature review with a perspective of genomic breeding for disease control in American mink (*Neogale vison*). Virus Research, 336, 199208. DOI Corresponding author (Review)

Mahmoodi, S. et al. (2023). Anthropogenic and natural fragmentations shape the spatial distribution and genetic diversity of roe deer in the marginal area of its geographic range. Ecological Indicators, 154, 110835. DOI Corresponding author

Vahedi, S. M. et al. (2023). Expanding the application of haplotype-based genomic predictions to the wild: A case of antibody response against *Teladorsagia circumcincta* in Soay sheep. BMC Genomics, 24, 335. DOI Corresponding author

Naserkheil, M. et al. (2022). Multi-omics integration and network analysis reveal potential hub genes and genetic mechanisms regulating bovine mastitis. Current Issues in Molecular Biology, 44(1), 309-328. DOI Co-author

Nazari, F. et al. (2022). A genome-wide scan for signatures of selection in Kurdish horse breed. Journal of Equine Veterinary Science, 113, 103916. DOI Co-author

Vahedi, S. M. et al. (2022). Weighted single-step GWAS for body mass index and scans for recent signatures of selection in Yorkshire pigs. Journal of Heredity, 113(3), 325-335. DOI Co-author

Taherkhani, L. et al. (2022). The candidate chromosomal regions responsible for milk yield of cow: a GWAS meta-analysis. Animals, 12(5). DOI Corresponding author

Golshani Jourshari, M. et al. (2022). Principal component analysis of biometric traits in Guilan native cattle of Iran. Iranian Journal of Applied Animal Science, 12(1), 23-31. Co-author

Vahedi, S. M. et al. (2022). Genome-wide selection signatures and human-mediated introgression events in *Bos taurus indicus*-influenced composite beef cattle. Frontiers in Genetics, 13, 844653. DOI Co-author

Varkoohi, S. et al. (2021). Allele Specific Expression (ASE) analysis between *Bos taurus* and *Bos indicus* cows using RNA-Seq data at SNP level and gene level. Anais da Academia Brasileira de Ciências, 93(3), e20191453. DOI Co-author

Salek Ardestani, S. et al. (2020). Signatures of selection analysis using whole-genome sequence data reveals novel candidate genes for pony and light horse types. Genome, 63(8), 387-396. DOI Co-author

Lado, S. et al. (2020). Genome-wide diversity and global migration patterns in dromedaries follow ancient caravan routes. Nature Communications Biology, 3, 387. DOI Co-author

Salek Ardestani, S. et al. (2019). Whole-genome signatures of selection in sport horses revealed selection footprints related to musculoskeletal system development processes. Animals, 10(1). DOI Co-author

Almathen, F. et al. (2016). Ancient and modern DNA reveal dynamics of domestication and cross-continental dispersal of the dromedary. Proceedings of the National Academy of Sciences (PNAS), 113(24), 6707-6712. DOI Co-author

Banabazi, M. H. et al. (2016). SNP calling in transcriptome of Holstein cows and their contribution in genetic variance of residual feed intake. Journal of Animal Science, 94(supp_4), 15. DOI First author

Chessa, B. et al. (2009). Revealing the history of sheep domestication using retrovirus integrations. Science, 324(5926), 532-536. DOI Co-author

Esmaeilkhanian, S., & Banabazi, M. H. (2006). Genetic variation within and between five Iranian sheep populations using microsatellite markers. Pakistan Journal of Biological Sciences, 9(13), 2488-2492. DOI Corresponding author

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